

ECS 408/608

Assignment 3

Interprocess Communication

Solutions

Kunwar Arpit Singh, 22185

For code repository of this assignment: [📁Operating Systems Assignments](#)

1. Problem 1

Problem statement

Write C program using IPC-Shared Memory that performs following two task using producer-consumer paradigm:

- (a) Producer (producer.c): Ask the user for an integer and string and place them in shared memory.
- (b) Consumer (consumer.c): Print the string input in the producer and check if the input number is even or not.

Answers

1.1 Code:

Shared:

```
1 #ifndef SHARED_H
2 #define SHARED_H
3
4 struct shared_data {
5     int number;
6     char text[100];
7 };
8
9 #endif
```

Producer:

```
1 #include <stdio.h>
2 #include <sys/ipc.h>
3 #include <sys/shm.h>
4 #include <string.h>
5 #include "shared.h"
6
7 int main() {
8
9     key_t key = 1234;
```

```

11     int shmid = shmget(key, sizeof(struct shared_data), 0666 | IPC_CREAT);
12
13     struct shared_data *data = (struct shared_data*) shmat(shmid, NULL, 0);
14
15     printf("Enter an integer: ");
16     scanf("%d", &data->number);
17
18     printf("Enter a string: ");
19     scanf("%s", data->text);
20
21     printf("Data written to shared memory\n");
22
23     shmdt(data);
24
25     return 0;
26 }

```

Consumer:

```

1  #include <stdio.h>
2  #include <sys/ipc.h>
3  #include <sys/shm.h>
4  #include "shared.h"
5
6  int main() {
7
8      key_t key = 1234;
9
10     int shmid = shmget(key, sizeof(struct shared_data), 0666);
11
12     struct shared_data *data = (struct shared_data*) shmat(shmid, NULL, 0);
13
14     printf("String from producer: %s\n", data->text);
15
16     if(data->number % 2 == 0)
17         printf("Number %d is EVEN\n", data->number);
18     else
19         printf("Number %d is ODD\n", data->number);
20
21     shmdt(data);
22
23     shmctl(shmid, IPC_RMID, NULL);
24
25     return 0;
26 }

```

1.2 Output:

a)

```

1 Enter an integer: 5
2 Enter a string: abc

```

```
3 Data written to shared memory
4
5 String from producer: abc
6 Number 5 is ODD
```

b)

```
1 Enter an integer: 4
2 Enter a string: abcd
3 Data written to shared memory
4
5 String from producer: abcd
6 Number 4 is EVEN
```

Output Screenshots

Terminal output

```
kas4453@LAPTOP-BU4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem1_producer.c -o assign3_problem1_producer
kas4453@LAPTOP-BU4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem1_consumer.c -o assign3_problem1_consumer
kas4453@LAPTOP-BU4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_producer
Data written to shared memory
String from producer: abc
Number 5 is ODD
kas4453@LAPTOP-BU4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem1_consumer
Data written to shared memory
String from producer: abcd
Number 4 is EVEN
kas4453@LAPTOP-BU4EH377:/mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$
```

2. Problem 2

Problem statement

Write a C program using IPC-Shared Memory that perform the following task using producer-consumer paradigm:

- (a) Producer (producer.c): Generate 100 random numbers and place them in shared memory.
- (b) Consumer (consumer.c): Read all the 100 numbers and generate the cumulative sum and exit.

Answers

2.1 Code:

Producer:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <sys/ipc.h>
4 #include <sys/shm.h>
5 #include <time.h>
6
7 #define SIZE 100
8
9 int main() {
10
11     key_t key = 5678;
12
13     int shmid = shmget(key, SIZE * sizeof(int), 0666 | IPC_CREAT);
14
15     int *numbers = (int*) shmat(shmid, NULL, 0);
16
17     srand(time(NULL));
18
19     printf("Generated_Numbers:\n");
20
21     for(int i=0;i<SIZE;i++)
22     {
23         numbers[i] = rand() % 100;
24         printf("%d_", numbers[i]);
25     }
26
27     printf("\nNumbers_written_to_shared_memory\n");
28
29     shmdt(numbers);
30
31     return 0;
32 }
```

Consumer:

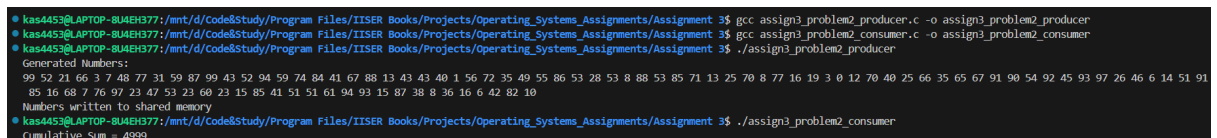
```
1 #include <stdio.h>
2 #include <sys/ipc.h>
3 #include <sys/shm.h>
4
5 #define SIZE 100
6
7 int main() {
8
9     key_t key = 5678;
10
11     int shmid = shmget(key, SIZE*sizeof(int), 0666);
12
13     int *numbers = (int*) shmat(shmid, NULL, 0);
14
15     int sum = 0;
16
17     for(int i=0;i<SIZE;i++)
18         sum += numbers[i];
19
20     printf("Cumulative Sum = %d\n", sum);
21
22     shmdt(numbers);
23
24     shmctl(shmid, IPC_RMID, NULL);
25
26     return 0;
27 }
```

2.2 Output:

```
1 Generated Numbers:
2 99 52 21 66 3 7 48 77 31 59 87 99 43 52 94 59 74 84 41 67 88 13 43 43 40 1 56
   72 35 49 55 86 53 28 53 8 88 53 85 71 13 25 70 8 77 16 19 3 0 12 70 40 25
   66 35 65 67 91 90 54 92 45 93 97 26 46 6 14 51 91 85 16 68 7 76 97 23 47 53
   23 60 23 15 85 41 51 51 61 94 93 15 87 38 8 36 16 6 42 82 10
3 Numbers written to shared memory
4
5 Cumulative Sum = 4999
```

Output Screenshots

Terminal output



```
kas4453@LAPTOP-8U4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem2_producer.c -o assign3_problem2_producer
kas4453@LAPTOP-8U4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem2_consumer.c -o assign3_problem2_consumer
kas4453@LAPTOP-8U4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem2_producer
Generated Numbers:
99 52 21 66 3 7 48 77 31 59 87 99 43 52 94 59 74 84 41 67 88 13 43 43 40 1 56
85 16 68 7 76 97 23 47 53 23 60 23 15 85 41 51 51 61 94 93 15 87 38 8 36 16 6 42 82 10
Numbers written to shared memory
kas4453@LAPTOP-8U4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem2_consumer
Cumulative Sum = 4999
```

3. Problem 3

Problem statement

Write a C program using IPC-Shared Memory that perform the following tasks:

- (a) Use shared memory to store an array of numbers.
- (b) The parent process writes 10 numbers into the shared memory.
- (c) Create 3 consumer processes where:
 - i. Consumer 1 computes the factorial of the first four numbers.
 - ii. Consumer 2 computes the factorial of the next two numbers.
 - iii. Consumer 3 computes the remaining numbers.

Answers

3.1 Code:

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <sys/ipc.h>
4 #include <sys/shm.h>
5 #include <unistd.h>
6 #include <sys/wait.h>
7
8 #define SIZE 10
9
10 long factorial(int n)
11 {
12     long f = 1;
13     for(int i = 1; i <= n; i++)
14         f *= i;
15     return f;
16 }
17
18 int main()
19 {
20     key_t key = 5678;
21
22     int shmid = shmget(key, SIZE * sizeof(int), 0666 | IPC_CREAT);
23
24     int *arr = (int *)shmat(shmid, NULL, 0);
25
26     printf("Enter 10 numbers:\n");
27
28     for(int i = 0; i < SIZE; i++)
29         scanf("%d", &arr[i]);
30
31     /* Consumer 1 */
```

```

32     if(fork() == 0)
33     {
34         printf("\nConsumer_1_(first_4_numbers)\n");
35         for(int i = 0; i < 4; i++)
36             printf("Factorial_of_%d=%ld\n", arr[i], factorial(arr[i]));
37         exit(0);
38     }
39
40     /* Consumer 2 */
41     if(fork() == 0)
42     {
43         printf("\nConsumer_2_(next_2_numbers)\n");
44         for(int i = 4; i < 6; i++)
45             printf("Factorial_of_%d=%ld\n", arr[i], factorial(arr[i]));
46         exit(0);
47     }
48
49     /* Consumer 3 */
50     if(fork() == 0)
51     {
52         printf("\nConsumer_3_(remaining_numbers)\n");
53         for(int i = 6; i < 10; i++)
54             printf("Factorial_of_%d=%ld\n", arr[i], factorial(arr[i]));
55         exit(0);
56     }
57
58     /* wait for children */
59     for(int i = 0; i < 3; i++)
60         wait(NULL);
61
62     shmdt(arr);
63     shmctl(shmid, IPC_RMID, NULL);
64
65     return 0;
66 }

```

3.2 Output:

```

1 Enter 10 numbers:
2 4 9 10 3 1 3 15 5 2 8
3
4 Consumer 1 (first 4 numbers)
5 Factorial of 4 = 24
6 Factorial of 9 = 362880
7 Factorial of 10 = 3628800
8 Factorial of 3 = 6
9
10 Consumer 2 (next 2 numbers)
11 Factorial of 1 = 1
12 Factorial of 3 = 6

```

```
13
14 Consumer 3 (remaining numbers)
15 Factorial of 15 = 1307674368000
16 Factorial of 5 = 120
17 Factorial of 2 = 2
18 Factorial of 8 = 40320
```

Output Screenshots

Terminal output

```
kas4453@LAPTOP-BU4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ gcc assign3_problem3.c -o assign3_problem3
kas4453@LAPTOP-BU4EH377: /mnt/d/Code&Study/Program Files/IISER Books/Projects/Operating_Systems_Assignments/Assignment 3$ ./assign3_problem3
Enter 10 numbers:
4 9 10 3 1 3 15 5 2 8

Consumer 1 (first 4 numbers)
Factorial of 4 = 24
Factorial of 9 = 362880
Factorial of 10 = 3628800
Factorial of 3 = 6

Consumer 2 (next 2 numbers)
Factorial of 1 = 1
Factorial of 3 = 6

Consumer 3 (remaining numbers)
Factorial of 15 = 1307674368000
Factorial of 5 = 120
Factorial of 2 = 2
Factorial of 8 = 40320
```